

MaskBeat: Loopable Drum Beat Generation

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Overview

- **MaskBeat** introduces a novel transformer-based architecture inspired by MaskGIT, using masked token prediction with bidirectional attention and iterative refinement. This allows simultaneous generation of drum instruments while maintaining groove consistency and musical coherence.
- The model incorporates custom loss functions: **groove loss**, **instrument dependency loss**, and **focal loss**. These losses encode domain-specific knowledge about drum patterns, significantly improving rhythmic structure and stylistic fidelity over standard losses.
- Trained on 30k high-quality loops, MaskBeat outperforms autoregressive and baseline MaskGIT models across beat strength, pattern repetition, and instrument balance metrics, while also enabling user control via an **interactive web app** with lockable instruments and a creativity slider.

MaskGit

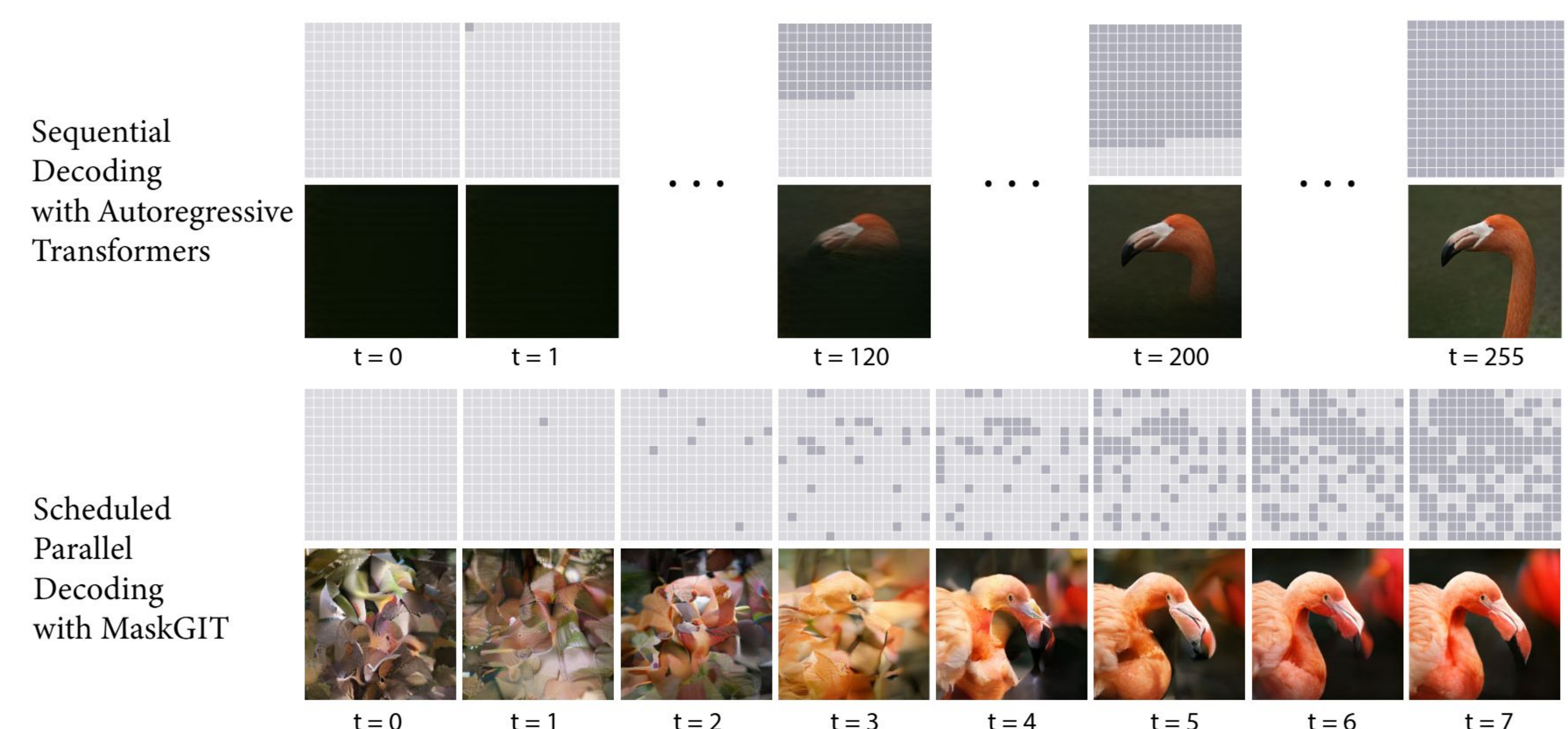


Image taken from Chang, Huiwen, et al. "Maskgit: Masked generative image transformer." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022.

Losses

$$\mathcal{L}_{\text{dep}} = \lambda_{\text{ks}} \sum_i [(1 - \hat{y}_{i,\text{kick}}) + (1 - \hat{y}_{i+4,\text{snare}})] + \lambda_{\text{hh}} \sum_t \hat{y}_{t,\text{CHH}} \hat{y}_{t,\text{OHH}} + \lambda_{\text{tom}} \mathcal{L}_{\text{tom}}$$

$$\mathcal{L}_{\text{groove}} = \sum_t w_t \cdot \mathcal{L}_{\text{sub}}(t) + \beta \sum_m \|P_m - P_{m+1}\|_2^2$$

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log p_t$$

Detailed View of MaskBeat Web App

